Interactive Fiction

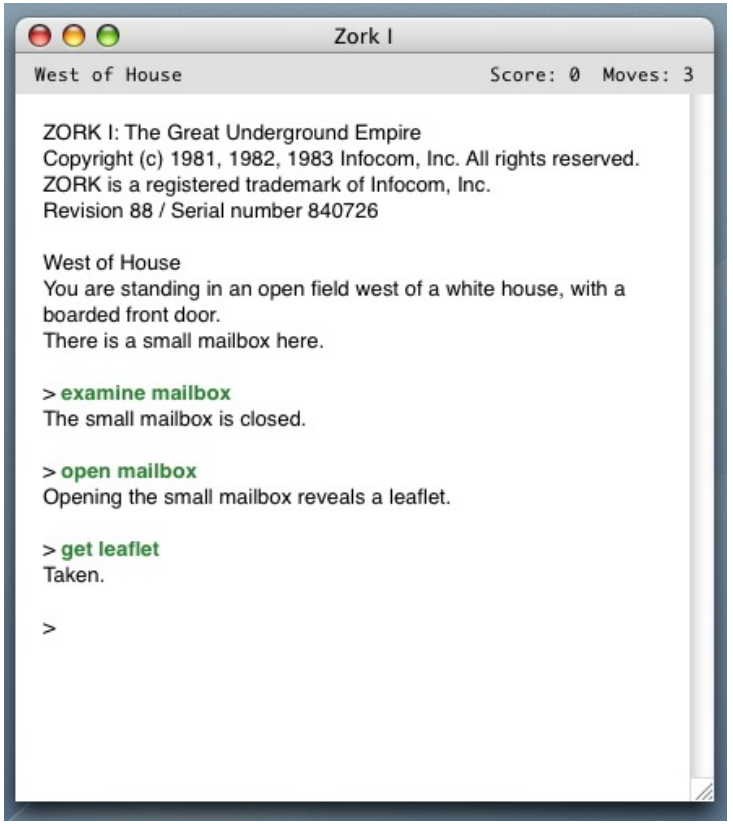
James Gain
Department of Computer Science
University of Cape Town
jgain@cs.uct.ac.za





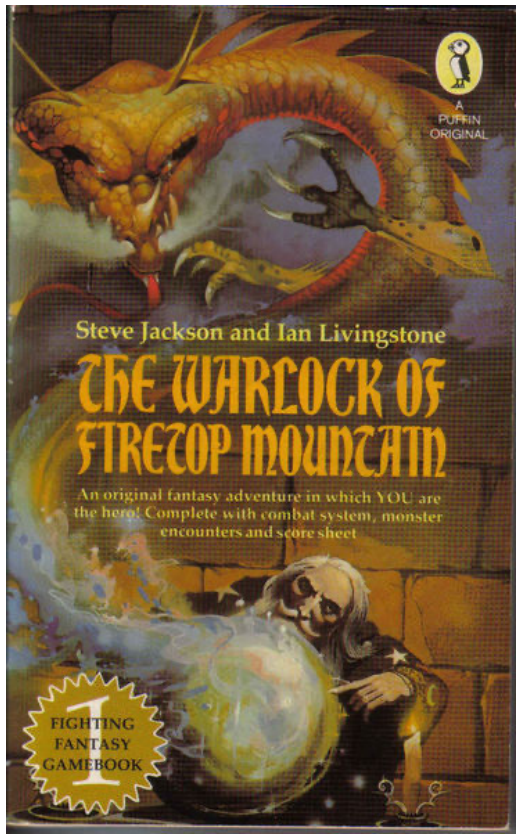
Interactive Fiction Defined

- Software environment in which players control a character with natural language text commands
- Characterised by text input and (mostly) text output
- Focuses on narrative and puzzles
- Input:
  - Simple imperative sentences
  - "Go West", "Examine Jar"
- Output:
  - Present tense, 2nd person viewpoint



Flavours

- Text Adventures
  - Purest and earliest form of IF
  - Used to distinguish from flavours such as graphic adventures
- MUDs (Multi-User Dungeons)
  - Multi-player internet game with elements of RPGs and instant messaging
  - A kind of multi-user or collaborative IF
- Choose Your Own Adventure
  - Book-based, where choices determine which section is read next
  - Fighting Fantasy series adds dice (much like tabletop RPGs)



How to Play: Getting Started

- Source: http://www.microheaven.com/IFGuide/
- Typically start with an intro:
  - Who you are, where you are, and (maybe) your objectives
- Interaction:
  - At the prompt (>) type imperative commands
  - *TAKE JAR*, *EXAMINE JAR*, *OPEN JAR* (articles can be skipped)
- Movement:
  - *GO SOUTH*, *SOUTH* or just *S*
  - Likewise for other directions, including *UP (U)* and *DOWN (D)*
  - Other ways to move around: *ENTER CAR, GO CAR, CLIMB ON SHIP, EXIT, LEAVE, DIVE INTO LAKE*
- If the game consistently refuses your commands then you are probably on the wrong track

How to Play: Common Commands

- Modern IF games recognize hundreds of verbs and even allow combinations
  - *PUT ALL BUT HAMMER IN SACK*
- For general game information, use *HELP* or *ABOUT*

LOOK (L)	*LOOK UNDER BED*
TAKE /DROP	*TAKE KNIFE*
LOCK / UNLOCK	*UNLOCK DOOR WITH RUSTY KEY*
EXAMINE (X)	*X KNIFE*
SEARCH	*SEARCH DESK*
INVENTORY (I)	*I*
OPEN / CLOSE	*OPEN DRAWER*
ASK / TELL / SAY	*ASK JOHN ABOUT POLICE OFFICER*
WAIT (Z)	*Z*
QUIT (Q)	*Q*

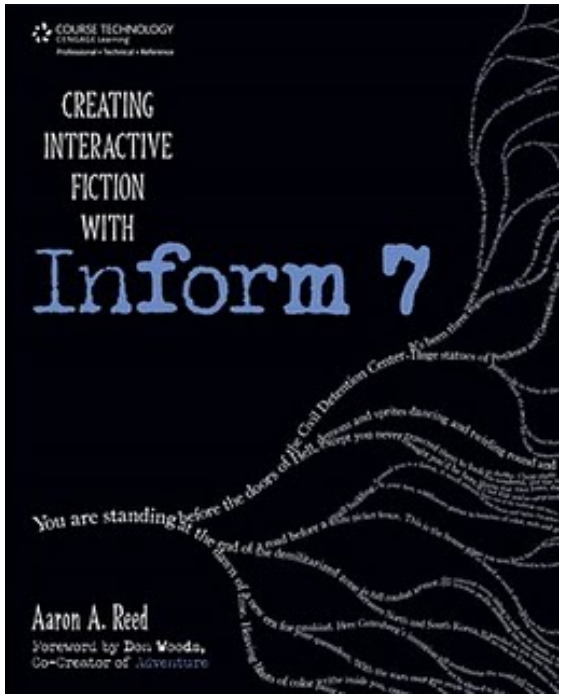
A Sample: Lost Pig

- "Lost Pig" by Admiral Jota
  - You are an orc called Grunk tasked with finding the farmer's pig
  - Pig is more intelligent than protagonist
  - Relatively short, but rich in responses
  - Genre: Fantasy, Humour
  - Winner, Best Game, Xyzzy Awards 2007
  - 8th place in IF Top 50 of all time



Inform 7

- A natural programming language and IDE for Interactive Fiction
  - Uses a simplified English-like syntax
  - Declarative rather than procedural
  - Powerful IF-specific debugging tools
- Arose in the modern era of community-supported IF
- Previous versions used a c-like syntax and were object-oriented



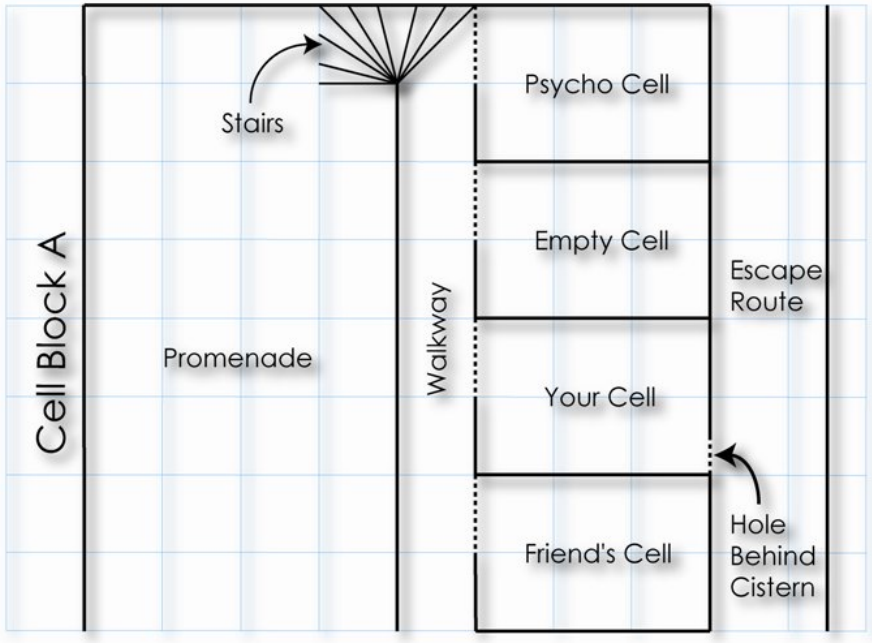
Inform7 Design Philosophy

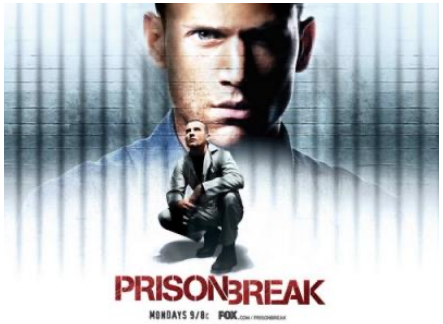
- "*The natural language in which to write interactive fiction is natural language*" – Graham Nelson

1. A casual reader should be able to correctly guess what a sentence does
2. The language should be economical but not at the expense of intelligibility
3. If in doubt as to syntax, imitate books or newspapers
4. Contextual knowledge is best supplied by the author, rather than being built in

Case Study: Prison Break

- Inspired by the TV series "Prison Break"
- Your task: to escape from Fox River Penitentiary







Case Study: Puzzle

- There is a hole behind the toilet Cistern in Your Cell
- But, the cistern can only be unfastened using a Sharpened Bolt on the Promenade floor
- However, the Psycho will attack you if you walk past his cell. So, you need to send your Friend to fetch the bolt
- Unfortunately, there are a limited number of turns before lockdown, when all the cell doors are closed
- Also, before unfastening the Cistern you will need to drape a Sheet so that the guards do not see your escape







Building the Prison

"Prison Break" by James Gain Your Cell is a room. "A steel-framed bunk with regulation sheets and a leaking toilet are squeezed into a space barely longer than a coffin. Carved into the wall is the motto: 'Michael was here!' The automatic gate to your cell is open to the west. Beyond the opening is an industrial steel walkway that overlooks a 'promenade' below." The Walkway is west of Your Cell. "Narrow. It's easy to slip and 'accidentally' fall over the railing. There are stairs at the far end, but Bagwell's cell is between you and them. 'Here fishy fishy fishy', he says." Test pb with "west / east".	**Prison Break** An Interactive Fiction by James Gain (with apologies to Fox International) **Your Cell** A steel-framed bunk with regulation sheets and a leaking toilet are squeezed into a space barely longer than a coffin. Carved into the wall is a motto: "Michael was here!" The automatic gate to your cell is open to the west. Beyond the opening is an industrial steel walkway that overlooks a "promenade" below. > test pb (Testing.) >[1] **west** **Walkway** Narrow. It's easy to slip and "accidentally" fall over the railing. There are stairs at the far end, but Bagwell's cell is between you and them. "Here fishy fishy fishy", he says. >[2] **east** **Your Cell**

Source Text

- General Program Structure:
  - Create things (using assertions in present tense to determine starting conditions)
  - With verbs – to be, to have, to carry, to wear, to contain, to support
  - Lay down rules (using instructions that modify state and behaviour)
  - Already has several hundred built-in rules
- Punctuation:
  - In general, normal punctuation applies
  - Full stops terminate commands
  - Square brackets ([ ]) represent comments or variables
  - Quotation marks (" ") are used for descriptions. Care needs to be taken for quotes within quotes

Prison Map

"Prison Break" by James Gain

Your Cell is a room. "..."

The Walkway is west of Your Cell. "..."

Psycho Cell is a room. North of the walkway is west of Psycho Cell.

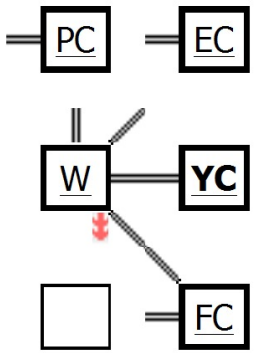
Empty Cell is a room. Northeast of the walkway is west of Empty Cell.

Friend's Cell is southeast of the Walkway. The Walkway is west of Friend's Cell.

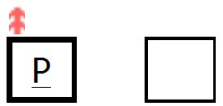
The Promenade is down from the walkway.

World

Map of upper level



Map of lower level

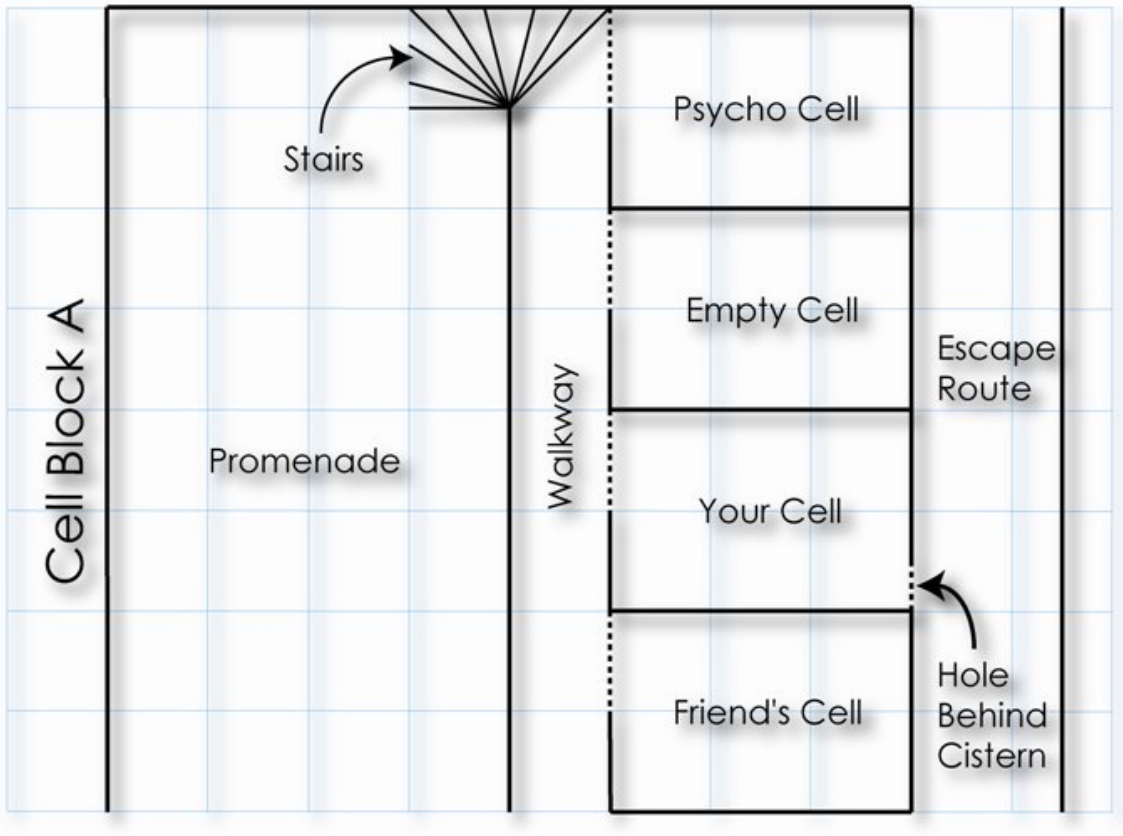


Mapping

- Maps are graphs:
  - Rooms are nodes, edges are connections
  - Connections are in the eight cardinal directions (north, northeast, etc.), as well as up and down
- Concept of inside and outside also supported:
  - "*Inside from the meadow is the woodcutter's hut*" (***from*** is NB here)
- Complex connections:
  - Connections are assumed symmetric
  - But this assumption can be overwritten - "*The Walkway is west of the Empty Cell. Your Cell is east of the Walkway*."
  - Can rewire connections to be asymmetric - "*North of the Walkway is west of Psycho Cell.*"
  - Use "nowhere" to create one-way connections - "*South of Psycho Cell is nowhere.*"

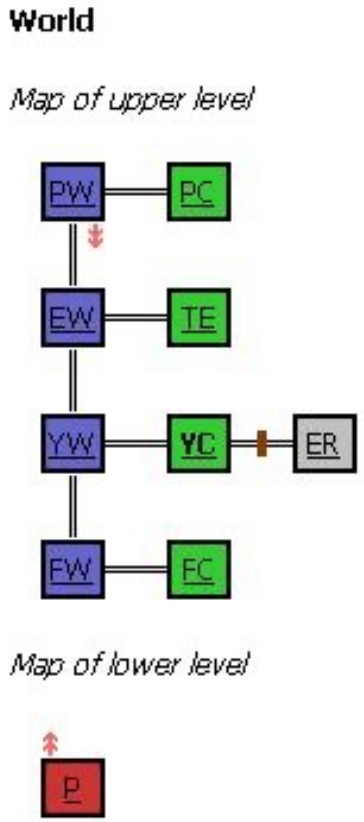
Mapping Exercise

- Use mapping assertions to create a more faithful representation of the prison



Regions

- Rooms can be grouped into regions
  - Regions can be placed inside one another but cannot overlap
  - Allows rules to be applied specifically to regions
  - Regions are colour-coded in the Inform7 World Index
- Example:
  - *AllCells is a region. Your cell, empty cell, friend's cell and psycho cell are in allcells. Allwalkways is a region. Your walkway, friend's walkway, empty walkway and psycho walkway are in allwalkways. Cell Block A is a region. Allcells, allwalkways and promenade are in Cell Block A.*



Kinds

- Object Types
  - room, thing, container, supporter, direction, etc.
  - Often assigned by inference from the text – *A book is on the library cart in your walkway*. (book is a thing, library cart is a supporter)
  - But can be assigned directly – *In your walkway is a supporter called the library cart*.
- Either/or Properties
  - Open/closed, openable/not openable, transparent/opaque, fixed in place/portable, edible/inedible
  - Can be assigned to objects of appropriate kind – *The Cistern is an openable opaque container. The Cistern is fixed in place.*

room
thing
  door
  container
    player's holdall
    vehicle
  supporter
  backdrop
  person
    animal
    man
    woman
  device
direction
region

Descriptions

- Objects can have three descriptions:
  1. On first appearance – *The bolt is here. "A long rusted bolt. It might serve as a weapon in an emergency."*
  2. When examined – *The description of the bolt is "On closer examination you see that the point has been filed slightly, as if somebody started making a shank but gave up."*
  3. In the inventory list – *The printed name of the bolt is "carefully-filed bolt".*



Doors

- Doors are interposed between rooms
  - *Your Cell Gate is west of Your Cell and east of Your Walkway. It is a door.*
  - Why do we need the 2nd sentence?
  - By default, doors have the property – fixed in place, closed and openable
- Locks
  - A door can be locked/unlocked, lockable/not lockable, have a matching key
  - *Your Cell Gate is lockable and locked. The Guards Key unlocks Your Cell Gate.*
  - Containers can also be locked but not supporters



Other Things

- Devices
  - Can be switched on or off (default) and descriptions changed accordingly
  - *The tape recorder is a device in your cell. "[if switched off] Devices like these are not allowed in prison. [otherwise] a voice on the tape says 'I'm sorry Michael. I dropped your bolt out on the promenade when the guards frisked me'."*
- Vehicles
  - An enterable container. Can be made pushable.
  - *In the yard is a vehicle called Prison Truck. It is pushable between rooms.*
  - Also extensions for vehicles that can be ridden on rather than in (horses, tricycles, etc.)
- Animate Beings
  - Person is a thing. Men, women and animals are kinds of person.
  - *In friend's cell is a man called Fernando Sucre.*



Exercise: Door to Freedom

- Detailing your Cell
  - The cistern, which gives access to the escape route and can only be moved with the bolt



Designing Interactive Fiction

- *An adventure game is a crossword at war with a narrative* - Graham Nelson
- Some Tips
  - Avoid linearity in narrative and puzzle-solving
  - Ensure internal consistency (realism)
  - Be descriptive but not overly so
  - Beware the trap of exponential growth
- The Player's Bill of Rights
  - Not to be killed without warning
  - Not to be given horribly unclear hints
  - To be able to win without experience of past lives
  - To be able to win without knowledge of future events
  - Not to have the game closed off without warning

Exercise: Solve “Icarus Dreams”

"Icarus Dreams" by James Gain

A ledge is a kind of room. The description of a ledge is “This ledge is narrow and crumbling. You press yourself flat against the cliff face to avoid falling into the sea below. You could go down or maybe east or west along the ledge.”

The Projecting Spur is a room. “A piece of rock projects out over the sea. It glistens in the sunlight.” Down from the spur is a ledge called High-ledge Left. Up from high-ledge left is nowhere.

East from high-ledge left is a ledge called High-ledge Right. The wax wings is here. The wax wings is wearable. The indefinite article of wax wings is “a pair of”.

Down from high-ledge right is a ledge called low-ledge right. Up from low-ledge right is nowhere. The climbing boots is here. The climbing boots is wearable. The indefinite article of climbing boots is “a pair of”.

Down from high-ledge left is a ledge called Low-ledge Left. Up from low-ledge left is nowhere. East of low-ledge left is a Low-ledge Right.

Down from low-ledge left is the Drowning Sea. Down from low-ledge right is the Drowning Sea. Nowhere is up from the drowning sea.

West of low-ledge left is the vertical crack. Up from the vertical crack is west of projecting spur. Vertical crack is a door. Vertical crack is locked and lockable. The matching key is the climbing boots.

Fly Away is a scene. Fly Away begins when play begins. Fly Away ends when the player is in the Projecting Spur and the player is wearing the wax wings.

When Fly Away ends:
  say “You run out along the projecting spur and fling yourself out into space. You plummet downwards but then an updraft catches your wings and you are borne up and away.”

